A PROJECT SYNOPSIS ON

SkillDNA

AI-Based Student Skill Assessment, Career Recommendation and Personalized Learning Roadmap

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering

Submitted by

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ABSTRACT

Engineering students often list many skills on their resumes, but they rarely know how strong each skill actually is or which career role they are closest to. Existing career portals give generic advice and resume screeners only check for keywords, so a student cannot see a clear, measurable picture of their abilities. **SkillDNA** is a web-based system that builds a "skill DNA" for every student by combining what they claim with what they can prove.

The student creates a profile, adds projects and uploads a resume in PDF form. The system extracts the resume text using PyPDF and uses a Large Language Model (LLM) served through the Groq API to detect the technical skills, a short profile summary and the projects mentioned. For each detected skill the system generates a fresh multiple-choice assessment of mixed difficulty. Each skill then receives a final score that combines the test result (70%) with resume and project evidence (30%), and is labelled Strong, Good, Developing or Weak.

Using this skill profile, the system recommends the five best-suited career roles with a readiness percentage, the matched and missing skills, and an average salary range in India. When the student selects a target career, SkillDNA performs a skill gap analysis and produces a personalised, phase-wise learning roadmap with topics, resources, durations and project ideas, and lets the student track progress step by step. The system is built with an HTML, Tailwind CSS and JavaScript frontend, a Python Flask backend and a MongoDB Atlas database.

Keywords: Skill Assessment, Resume Parsing, Natural Language Processing, Large Language Models, Career Recommendation, Skill Gap Analysis, Learning Roadmap.

1. INTRODUCTION

The gap between what students learn and what the industry expects is one of the main reasons fresh graduates struggle in placements. Students collect certificates and list technologies on their resumes, but self-reported skills are not a reliable measure of ability. At the same time, choosing a career path such as Data Analyst, Python Developer or Machine Learning Engineer is difficult without knowing which skills each role needs and how far the student is from it.

Recent advances in Natural Language Processing and Large Language Models make it possible to read a resume like a human reviewer, understand the skills it describes, and generate meaningful, skill-specific questions automatically. SkillDNA uses these capabilities to give each student an evidence-based skill profile, career suggestions based on that profile, and a concrete plan to close the remaining gaps.

a. Problem Statement

Students do not have a single tool that (i) verifies the skills they claim, (ii) measures the strength of each skill, (iii) maps their skill profile to suitable career roles, and (iv) tells them exactly what to learn next. Manual counselling does not scale, and keyword-based resume tools do not test real knowledge. There is a need for an intelligent system that connects resume analysis, skill testing, career matching and learning guidance in one flow.

b. Objectives

- To extract text from a student's resume PDF and automatically detect technical skills and projects using NLP and an LLM.
- To generate skill-specific multiple-choice assessments of mixed difficulty for the detected skills.
- To calculate a reliable score for every skill by combining assessment results with resume and project evidence, and classify each skill as Strong, Good, Developing or Weak.
- To recommend suitable career roles with a readiness percentage, matched skills and missing skills.
- To identify the skill gap for a chosen career role and generate a personalised, phase-wise learning roadmap.
- To let students track their progress on the roadmap through a simple web dashboard.

c. Scope

The system is aimed at undergraduate engineering students, mainly from Computer, IT and related branches, and focuses on technical skills such as programming languages, databases, data science, machine learning and web or app development. It is designed as a web application that can be used by an individual student or by a college training and placement cell.

2. LITERATURE SURVEY

Kashif and Kumar [1] present a resume parser based on NLP that extracts structured information such as skills, education and experience from unstructured resume text. Their work shows that rule-based and NLP-based extraction can convert free-form resumes into usable data, which is the first step of SkillDNA.

Wang, Su and Yu [2] study feature extraction and analysis of natural language for deep learning in English language learning. Their work highlights how language features can be extracted from text and used to assess and guide learners, which relates to the assessment and feedback part of this project.

Bhatia, Rawat, Kumar and Shah [3] propose an end-to-end resume parsing approach and use BERT to find suitable candidates for a job description. This shows that transformer-based language models can match a candidate profile to role requirements far better than plain keyword matching, which motivates the use of an LLM for career matching in SkillDNA.

Chikane et al. [4] describe an AI-based resume screening system using machine learning and natural language processing to rank resumes against job requirements. Such systems serve recruiters; SkillDNA applies a similar idea from the student's side, to show the student how ready they are for a role.

Research Gap

The surveyed systems mainly parse resumes or screen candidates for recruiters. They depend on what is written in the resume and do not test whether the candidate actually knows the skill. They also stop at a match score and do not give the learner a plan for improvement. SkillDNA fills this gap by adding skill verification through assessments, an evidence-weighted scoring model, and a personalised learning roadmap with progress tracking.

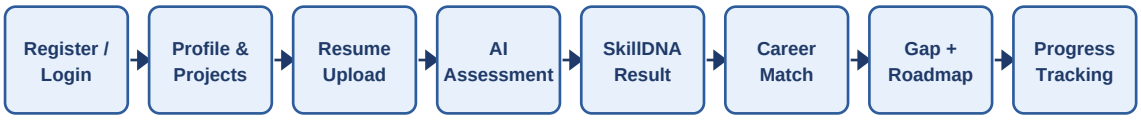
3. PROPOSED SYSTEM

a. Existing System

- Students judge their own skills, which is subjective and often inaccurate.
- Resume screening tools only match keywords and are built for recruiters, not for students.
- Online tests are generic and not linked to the student's resume or target career.
- Career guidance is manual, time-consuming and does not provide a structured learning plan.

b. Proposed System

SkillDNA connects every step of the student's journey into one flow:



- Resume skills are detected automatically and merged with the skills the student enters.
- Every skill is tested with AI-generated questions, so scores reflect real knowledge.
- Career readiness and skill gaps are based on measured scores, not only on claimed skills.
- The roadmap is personalised and the student can mark steps as completed to track progress.

c. Proposed Architecture

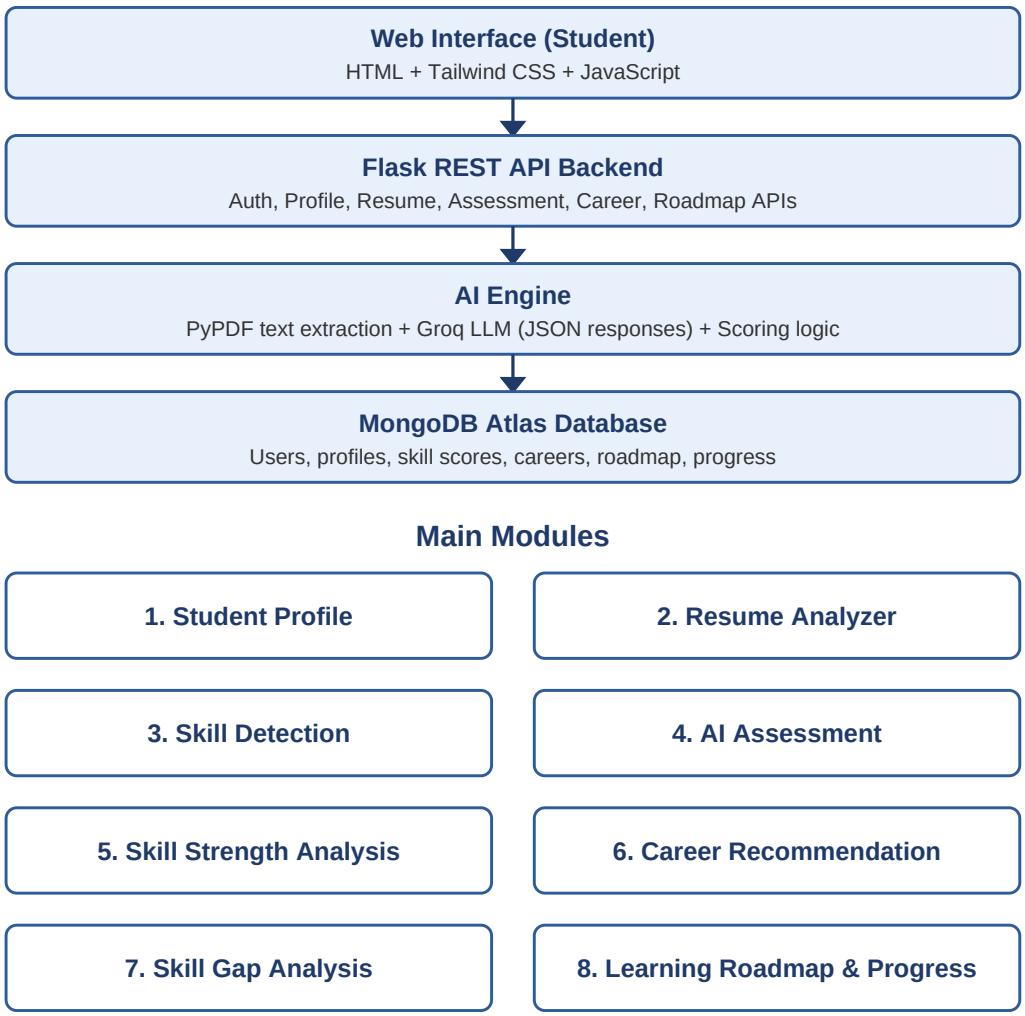


Figure 1: System architecture of SkillDNA

Module Description

Student Profile — The student registers and logs in securely (passwords are stored as hashes and requests use a session token). The student then enters education, branch, year, skills and project details, which form the base data used by every other module.

Resume Analyzer — The student uploads a resume in PDF form. The module extracts plain text from every page of the PDF using the PyPDF library and passes it to the AI engine.

Skill Detection — The extracted resume text is sent to the LLM, which returns the list of technical skills, a two-line profile summary and the projects mentioned, in JSON format. The detected skills are merged with the skills entered in the profile, removing duplicates.

AI Assessment — For up to five selected skills, the LLM generates exactly five multiple-choice questions per skill (2 easy, 2 medium, 1 hard), each with four options. Invalid questions are filtered out and the correct answers are kept on the server so they cannot be seen by the student.

Skill Strength Analysis — After submission, the module calculates a test score and an evidence score for each skill, combines them into a final score, and labels the skill as Strong, Good, Developing or Weak. A question-wise review with correct answers is also shown.

Career Recommendation — The student's skill scores, other skills, projects and resume summary are given to the LLM, which recommends the five best career roles with a readiness percentage, a reason, matched skills, missing skills and an average salary range in India, sorted by readiness.

Skill Gap Analysis — For the career role chosen by the student, the system compares the required skills with the student's current skills and marks each one as have, partial or missing, with a short note.

Learning Roadmap & Progress — The identified gaps are arranged into 6–8 ordered steps, starting from weak areas and ending with portfolio, resume and interview preparation. Each step has a duration, topics, learning resources and a project idea. The student can mark steps as completed and the progress is saved.

d. Scoring Methodology

Each assessed skill receives a final score out of 100 that combines what the student can prove in the test with the evidence of that skill in their resume and projects:

Test Score = (correct answers / total questions) × 100

Evidence Score = 20 + 40 (if the skill appears in the resume) + 40 (if the skill appears in a project)

Final Score = 0.7 × Test Score + 0.3 × Evidence Score

Final Score	Skill Level	Meaning
80 – 100	Strong	Job-ready in this skill
60 – 79	Good	Solid base, needs some practice
40 – 59	Developing	Basic knowledge, needs focused learning
Below 40	Weak	Should be learnt from the fundamentals

Giving 70% weight to the test keeps the score based on real knowledge, while the 30% evidence weight rewards students who have actually applied the skill in their resume and projects.

e. Experimental Setup

1. Software Requirements

Operating System: Windows 10/11 or Linux

Programming Language: Python (backend), JavaScript (frontend)

Development Environment: Visual Studio Code

Frontend: HTML, Tailwind CSS, JavaScript

Backend: Flask (Python)

Database: MongoDB Atlas (cloud)

AI Service: Groq API with a Large Language Model (default: openai/gpt-oss-120b)

Browser: Google Chrome / Microsoft Edge / Mozilla Firefox

2. Hardware Requirements

Processor: Intel Core i5 or equivalent

RAM: Minimum 8 GB

Storage: Minimum 128 GB SSD

Display: Standard laptop or monitor

Internet Connection: Required for the Groq AI API and the MongoDB Atlas database

GPU: Not required (AI processing runs on the cloud)

3. Libraries

Flask: Backend REST API development and serving the web interface

Flask-CORS: Handling cross-origin requests between frontend and backend

PyPDF: Extracting text from resume PDFs

Groq SDK: Calling the LLM for skill extraction, question generation, career recommendation and roadmap generation

PyMongo / dnspython: Connecting to the MongoDB Atlas database

Werkzeug Security: Password hashing and verification

python-dotenv: Loading API keys and database settings from the environment file

4. Dataset

A small set of sample resumes from different branches and skill levels will be used to develop and test the resume analyzer and skill detection modules.

Assessment questions are not taken from a fixed question bank; they are generated by the LLM for each student and each skill, covering technical skills such as Python, SQL, Data Science and Machine Learning, so the same student gets different questions in each attempt.

Skill requirements for career roles such as Python Developer, Data Analyst, Web Developer and Machine Learning Engineer come from the LLM's knowledge of industry roles and are compared with the student's measured skill scores.

The system will be tested on the accuracy of skill detection from resumes, the correctness of the generated questions and calculated skill scores, the relevance of career recommendations, and the usefulness of the generated learning roadmap.

4. EXPECTED OUTCOMES

- Automatic detection of skills and projects from an uploaded resume.
- A verified, score-based skill profile (“SkillDNA”) for every student.
- Top five career recommendations with readiness percentage and missing skills.
- A personalised, step-by-step learning roadmap with resources and project ideas.
- A dashboard where students can track roadmap progress over time.

5. APPLICATIONS

- Self-assessment and placement preparation for engineering students.
- Training and placement cells to understand the skill level of a batch.
- Faculty mentors to guide students towards suitable career paths.
- Coaching institutes and online learning platforms for personalised learning plans.

6. FUTURE SCOPE

- Coding and practical tasks in addition to multiple-choice questions.
- Soft-skill and aptitude assessment.
- Live job-market data to keep career requirements and salary ranges up to date.
- Admin dashboard for colleges with batch-level analytics.
- Mobile application version of the system.

7. CONCLUSION

SkillDNA provides students with a clear, evidence-based view of their technical abilities. By combining resume analysis, AI-generated assessments, a weighted scoring model, career matching and personalised roadmaps in a single web application, the system helps students understand where they stand, which career suits them best and exactly what to learn next. This makes career planning more accurate, measurable and goal-oriented than self-assessment or generic guidance.

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